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Northward displacement of north-central British Columbia

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An extensive terrain in north-central British Columbia, apparently moved ~1,300 km northwards along the western fringe of North America during the late Cretaceous or early Tertiary, perhaps in much the same fashion as Baja California is doing today. Large anticlockwise rotations occurred within this terrain.

THE western, volcanosedimentary 'eugeosyncline' of the North American Cordillera is a collage of terrains of varied origins that apparently became attached to the western continental margin of the continent at different times[1-3]. These terrains now lie to the west of the line WW in Fig. 1, which is the probable western limit of the ancient North American craton. It is delincated by the westernmost limit of Palaeozoic and Precambrian strata whose continuity with the craton can be reasonably inferred, and by the transition of the $^{87}Sr/^{86}Sr$ initial ratios from <0.704 to the west (indicating that the underlying crust is of oceanic or island arc origin) to >0.706 to the east (indicative of continental crust)[4]. EE is the eastern limit of Cordilleran deformation.

The concept of a tectonic collage derives from several lines of evidence. Each terrain has an early Mesozoic and late Palaeozoic stratigraphy which differs from that of contiguous terrains[5,6], each has early Mesozoic and late Palaeozoic faunas that differ from coeval faunas along the ancient cratonic margin[7,8], and several have yielded palaeomagnetic results indicating that they have been displaced latitudinally or rotated by large angles about local vertical axes[9-12]. The Alaskan peninsula and the Chugach Mountains (SAJ of Fig. 1), composed of Jurassic and younger rocks, have been displaced 1,500 km from the south since the Jurassic[10]. A western terrain[13], sometimes called Wrangellia[5], with distinctive Triassic stratigraphy, comprises parts of southeastern Alaska, Queen Charlotte Islands, Vancouver Island and the Hell's Canyon area of eastern Oregon the latter denoted by '?' in Fig. 1*b*. Wrangellia has moved at least 1,500 km, and possibly as much as 5,000 km, from the south since the late Triassic[9,11]. Much of southeastern Alaska is underlain by the Alexander terrain, with characteristic Palaeozoic and Triassic strata[14]. Recent palaeomagnetic results[15] from the Alexander terrain indicate large anticlockwise rotations and yield Triassic palaeolatitudes consistent with those for the craton. Instances of northward displacements and clockwise rotations have been reported from batholithic rocks[16,17]. The Coast Range of Oregon[18,19], and the Transverse Ranges of southern California[20] have rotated clockwise by ~60° during the Tertiary. We now present palaeomagnetic evidence from a reconnaissance survey for the northwards displacement and differential rotation of an extensive tract of country in northern British Columbia known as the Stikine terrain, together with rocks of the Cache Creek Group that lie along its eastern border.

The integrity of the Stikine terrain during the early Mesozoic is shown by the distinctive late Triassic and early to middle Jurassic rocks that extend across it. It is bordered to the east by the Cache Creek Group, which lies just west of WW. The Cache Creek Group is interpreted as late Palaeozoic–early Mesozoic ancestral Pacific Ocean crust with its overlying deposits that were thrust westwards on to the Stikine terrain in post-Oxfordian time[2,6,21]. To the east of WW there are volcanogenic terrains that may have formed along or immediately offshore from the Pacific margin of North America during the Mesozoic[2]. The displacement of terrains in the western Cordillera from the south has been suggested first to account for the juxtaposition of warmer-water western Triassic faunas and cooler-water faunas in the Rocky Mountains[7], and second to explain the peculiar distribution of Permian fusulinids whereby the Cache Creek Group, with its distinctive verbeekinid or Tethyan faunas, was inserted between schwagerinid or American faunas that occur to the west in the Stikine terrain and to the east and south in south-central British Columbia, Oregon and California[8]. Our palaeomagnetic studies were undertaken to make quantitative estimates of any displacements that might have occurred.

Palaeomagnetic results

The rocks sampled are Upper Triassic and Lower Jurassic volcanogenic strata, typical of the Stikine terrain, and a Lower Cretaceous layered intrusion cutting deformed Cache Creek rocks (Fig. 2). These three rock units were sampled at 42 sites distributed among six localities. Within each locality the sequences are continuous, with no field evidence of relative motion. Attitudes of bedding, flow layering or compositional layering were measured at each site, with the exception of some massive augite porphyries at the base of the Triassic section (Table 1, section (7)), but their continuity with overlying, compositionally similar, uniformly dipping flows suggests they have the same attitudes. Three to six oriented samples (cores or hand samples) were obtained from each site. Two cores were cut from each hand sample, making a total of 236 cores. Two specimens were cut and measured from each core.

The oldest rocks studied are from the Savage Mountain Formation of the Takla Group of late Carnian to earliest Norian age[22]. The Savage Mountain Formation consists of up to 5,000 m of dark grey-green, locally red, pillow basalt and massive subaerial flows, with associated volcanoclastics. Samples were obtained from two localities, mainly from subaerial flows. At all collecting sites, with the exception noted above, attitudes could be accurately measured. When sampling

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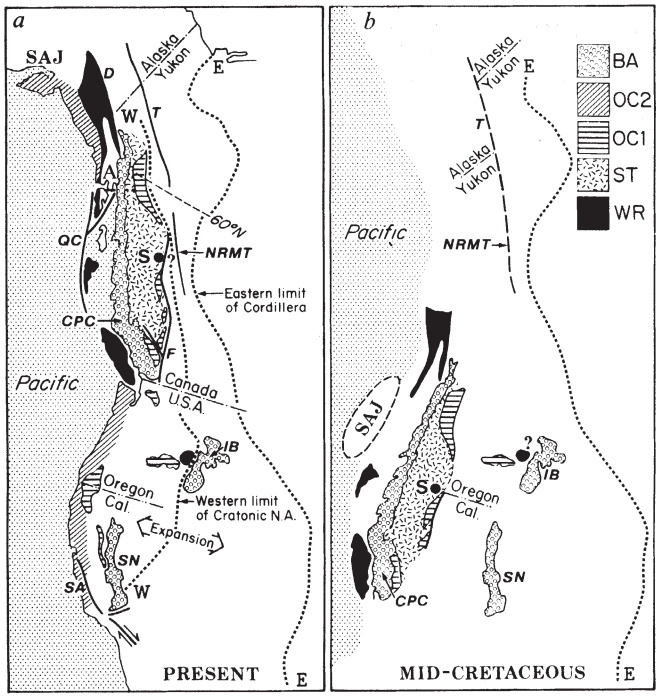


Fig. 1 *a*, A sketch-map showing some of the major tectonic elements of the Cordillera. EE is the easterly limit of deformation; WW, the westerly limit of the North American craton. BA, major plutons (CPC, Coast Plutonic Complex; IB, Idaho batholith; SN, Sierra Nevada batholith); WR, Wrangellia; ST, Stikine terrain; OC1, mid-Jurassic and older oceanic crust and rocks deposited on it; OC2, late Mesozoic and Tertiary oceanic crust and rocks deposited on it; A, the Alexander terrain is the blank area. *b*, The proposed position of the Stikine and Wrangellian terrains during the late Triassic to mid-Cretaceous. The Tertiary expansion of the Basin and Range Province is restored, and Wrangellia and the Coast Plutonic Complex are attached to the Stikine terrain in their present relative positions. These terrains are moved undistorted. We consider only gross latitudinal displacement, and no attempt is made to restore the original geometries of individual elements of the Stikine and Wrangellian terrains. Certainly large internal rotations are involved (see text). Major right-lateral strike-slip faults are marked: *D*, Denali; *F*, Frazer; *NRMT*, Northern Rocky Mountain Trench; *QC*, Queen Charlotte Island–Chatham Strait system; *SA*, San Andreas; *T*, Tintina. S is the sampling area, and SAJ is the late Jurassic displaced terrain of southern Alaska[10]. In *b* the Alaska–Yukon border is displaced on the Tintina–Northern Rocky Mountain Trench lineament to show the amount of motion that can be demonstrated geologically on one of the fault systems that might record some of the northward displacement of the Stikine terrain. The total displacement occurred by motions of comparable magnitude distributed among several such faults (see text).

Takla and Hazelton rocks, preference was given to deuterically oxidized basalts, because such rock types have been shown to be very stable magnetically[23], and our preliminary studies confirmed that this was also true for the sequences we have sampled (Fig. 3). Such oxidized rocks can usually be recognized by diagnostic minerals such as iddingsite in olivine basalts, because they are often reddish, and because they yield grey-red mud when drilled. The Takla basalts have alkaline affinities, and are generally porphyritic, with abundant clinopyroxene, feldspar, and altered olivine phenocrysts. The directions of natural remanent magnetizations (NRM) at each site are sometimes scattered, but are more commonly well grouped. During cleaning by alternating field (30–80 mT) or thermally (550–600 °C), they coalesce into good groups. The same directions persist after treatment in fields of 150 mT, and in some specimens, up to 200 mT. Blocking temperatures of this high coercive force magnetization are in the range 500–650 °C, and haematite and magnetite blocking temperatures often occur in the same specimen (Fig. 3). The directions remain unchanged between 500 and 650 °C, the haematite and magnetite magnetizations having identical directions. After cleaning, and after correction is made for the dip of the beds, the directions are westerly. This direction characterizes the Savage Mountain Formation in the places sampled. The two localities have statistically identical inclinations, but their mean declinations differ by 19° as if they have been rotated relative to one another about vertical axes (Fig. 4, Table 1). The mean inclination of 41° yields a palaeolatitude of 23°, which is significantly lower than the late Triassic palaeolatitudes calculated from cratonic North America (Fig. 5).

Lower Jurassic rocks of the Hazelton Group were sampled at three localities spread over 100 km (Fig. 2). The Hazelton Group consist of calc-alkaline volcanics and intercalated sediments. It ranges in age from late Sinemurian to early Callovian, and extends across the eastern and southern parts of the Stikine terrain[24]. The group contains three formations: Telkwa (bottom), Nilkitkwa and Smithers (top). Late Sinemurian subaerial red basalt, and red fine-grained tuff of the Telkwa Formation were sampled from the Red Creek–Two Lake Creek area, and the Driftwood Range. Toarcian basalt and red tuff of the Nilkitkwa Formation were obtained from the Bait Range. The directions of NRM at each collecting site are usually grouped together but are occasionally scattered. Cleaning by alternating fields and heating yields well-grouped directions. The coercivity commonly exceeds 100 mT, and blocking temperatures are in the range of 500 to 700 °C, except in some tuffs in which the range is 200–600 °C (Fig. 3). The directions at the sites within a locality agree reasonably well. Moreover, the mean inclinations are statistically identical from locality to locality, but the declinations range from south-west to north-northeast (Fig. 4, Table 1), as if the localities have been differentially rotated by large angles about vertical axes. Alternatively, one may argue that these directions (unrotated) could be the resultants of unresolved antiparallel magnetization acquired over a substantial time interval during lithification when the geomagnetic field reversed in direction. This is improbable

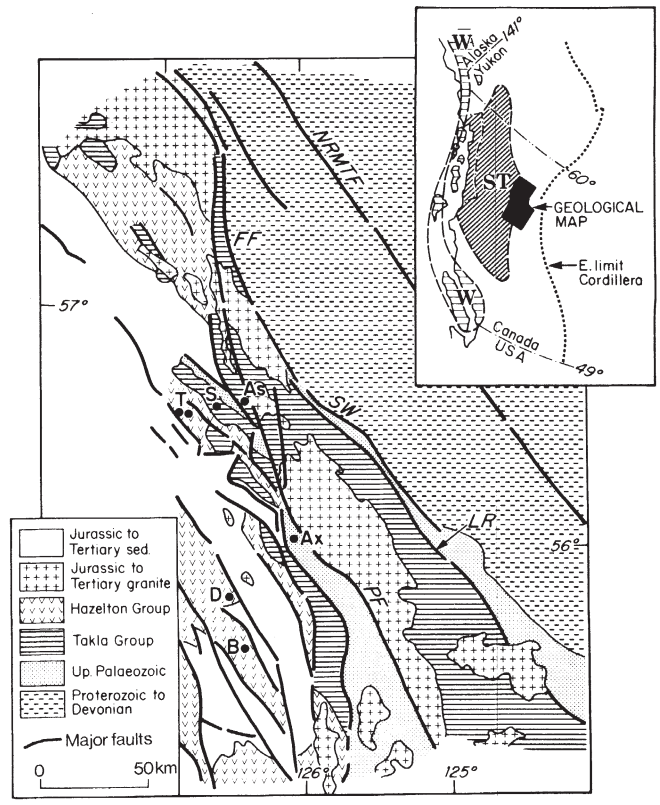


Fig. 2 Geology of the sampling region. The sampling localities are: As, Asitka Peak; Ax, Axelgold intrusion; B, Bait Range; S, Sustut Peak; and T, Two Lake Creek and Red Creek localities. The inset shows the general location and the two major displaced terrains, Wrangellia, W, and the Stikine terrain, ST. The faults marked are: FF, Findlay; LR, Lay Range; NRMT, Northern Rocky Mountain Trench; PF, Pinchi; and SW, Swannell.

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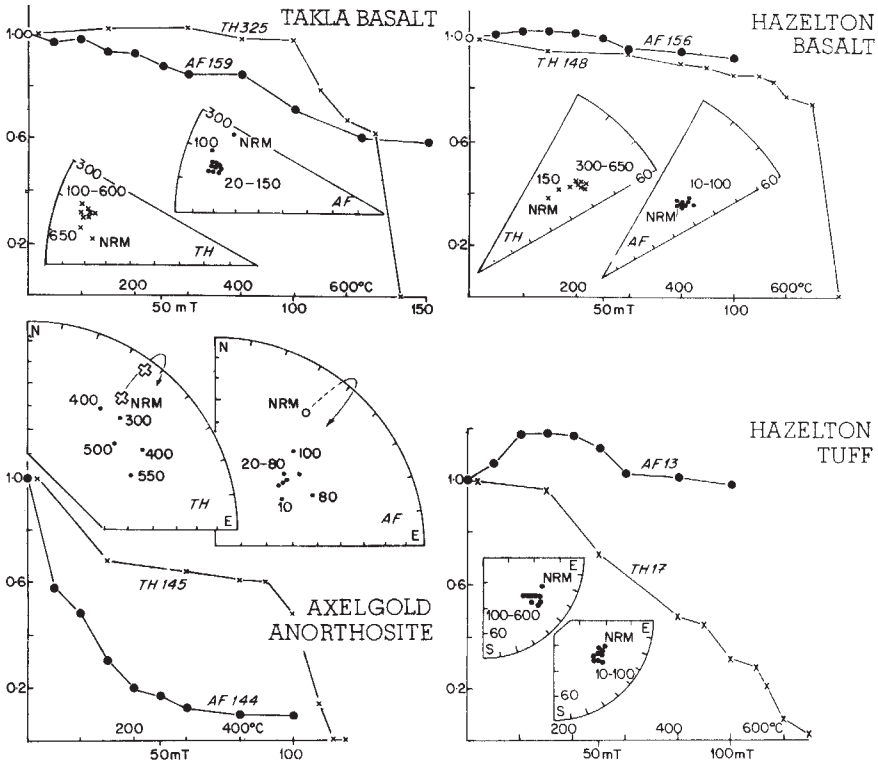


Fig. 3 Demagnetization studies. Examples of thermal (TH) and alternating field (AF) demagnetization of pairs of specimens from the same oriented core. Directions are plotted on equal-area projections, the plane is the present horizontal. The numbers on the demagnetization graphs are the intensities of natural remanent magnetization (NRM) in units of 10^{-3} A m^{-1}. The Takla and Hazelton basalts are deuterically oxidized; the directions are little changed throughout the demagnetization range; remanent coercive forces are high; blocking temperatures appropriate to haematite and fine-grained magnetite occur; both minerals can be identified in polished section and both have the same magnetization direction; inclinations are positive (downward). In the Axelgold anorthosite the directions initially have negative inclination and they migrate rapidly to the characteristic direction (inclination positive) for the intrusion during removal of a soft magnetization which is randomly directed from core to core and which we attribute to lightning; there is a well-marked 'magnetite' blocking temperature; in polished section magnetite occurs as very fine rods in felspar crystals; remanent coercive forces are typical of those of fine-grained magnetite; haematite magnetizations do not occur. The red tuff from the Hazelton Group has negative inclination and the direction is little changed through the demagnetization range; the magnetization is due to very fine-grained hematite readily observed in polished section.

because the haematite and magnetite magnetizations that probably formed at somewhat different times always have identical directions (Fig. 3). Moreover, it is highly improbable that such a complex magnetic structure would have produced inclinations of both normal and reversed magnetizations that are consistent from locality to locality. A further possibility is that the directions (unrotated) are the resultants of primary magnetizations and stable secondary magnetizations, the latter acquired long after formation. Again, such an interpretation is highly improbable for the two reasons just given, and because we have been careful never to sample close to later igneous bodies or zones of secondary mineralization. Therefore we conclude that large relative rotations have occurred. A mean inclination (54°) is obtained by rotating all site directions so that the mean declinations at each locality agree (Hazelton rotated, Fig. 4). The mean palaeolatitude of 35° is 14° farther south than expected, but the difference is marginally significant. The large error of Fig. 5 arises from the considerable uncertainty in the early Jurassic field direction for cratonic North America as it is presently known (see below). These results from the Hazelton Group provide one of the few well-documented instances of reversals of magnetization in early Jurassic rocks.

The third unit is the Axelgold Intrusion, a layered body, 12 by 4.5 km in extent[25]. Its age is less well defined than in the previous two units. The body intrudes highly deformed rocks of the Cache Creek Group, and cuts structures that, 25 km to the north, are congruent with structures of post-Oxfordian age. The intrusion is therefore post-Oxfordian. Two biotites and one hornblende from the reaction rim of a granite xenolith in the interior of the intrusion, and from a micaceous gabbro close to the contract, yield K–Ar ages in the range 98–108 Myr (ref. 25). A second hornblende gives a K–Ar age of 155 Myr, but is considered[25] to "reflect the presence of excess argon". A mid-Cretaceous age is the most probable[25], but an early Cretaceous or late Jurassic age cannot be discounted. A mid-Cretaceous age is also consistent with the fact that we have found only normal polarities in the body, and normal polarity typifies the geomagnetic field in the interval 82–110 Myr (ref. 26). If it were late Jurassic or early Cretaceous, both normal and reversed polarity would have been expected in such a large slowly-cooled body, because field reversals were then frequent[26].

The intrusion comprises alternating layers of anorthosite and anorthositic gabbro, ranging from ~10 to 50 cm in thickness, and dipping 10–15°. Anorthositic samples were chosen preferentially because preliminary studies showed them to be more stable magnetically. The layering was presumably produced by gravity-controlled magnetic currents and is therefore considered to be a good indicator of the palaeohorizontal. The NRM directions are usually scattered, but after cleaning in alternating magnetic fields they become well grouped, the characteristic magnetization having north-northeast declinations and steep downward inclinations (Fig. 4). The coercivity commonly exceeds 50 mT and the blocking temperatures are in the range of 500 to 600 C. The mean inclination relative to layering is 69°, yielding a palaeolatitude of 52°, about 13° lower than expected for the mid-Cretaceous. It is possible that the layering was not formed horizontally, or that tilting occurred before the acquisition of remanence whilst the body was hot, and this estimate of palaeolatitude would then be in error. Note that the directions of magnetization relative to present horizontal (Table 1) yield a palaeolatitude of 40° which is even more aberrant. A further source of error is in the age of the intrusion. If it were late Jurassic, then the estimate of palaeolatitude would be in good agreement with that expected for North America at that time (see ref. 12, Fig. 15), implying that both the Cache Creek terrain, into which the body is intruded and the Stikine terrain were then in place relative to cratonic North America. Thus it is important to note that our preferred interpretation of Fig. 5 is uncertain because of the difficulty in determining accurately the age and the palaeohorizontal.

Discussion

The results for these three time intervals all yield mean palaeolatitudes that are close to the latitude of the California–Oregon boundary. The Stikine terrain is geologically a coherent entity so that although the observations are from a geographically restricted area they, nevertheless, indicate that from the late Triassic until sometime in the Jurassic or

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Table 1 Summary of results

	Th	*B*	*n*	*D, I*	*k*	α_{95}°	Pole λ, ϕ	$(dm°, dp°)$
(1) Axelgold, 98–108 Myr	270	13	67	025°, +69°	70	05	76°N, 033°W	(08, 07)
(2) Axelgold, uncorrected	270	13	67	032°, +60°	92	04	64°N, 012°W	(07, 05)
(3) Hazelton, mid-Toarcian	170	07	37	359°, +55°	15	16	70°N, 057°E	(22, 16)
(4) Hazelton, Sinemurian	070	04	28	242°, +56°	29	18	17°N, 175°W	(25, 18)
(5) Hazelton, Sinemurian	125	04	20	114°, −52°	14	25	40°N, 144°E	(35, 24)
(6) Takla, late Triassic	150	06	42	300°, +44°	31	06	38°N, 133°E	(15, 10)
(7) Takla, late Triassic	200	08	42	281°, +38°	18	07	24°N, 146°E	(16, 09)

(1) Axelgold layered intrusion, Axelgold Range 56.2°N, 126.1°W, cleaning 30–60 mT, normal polarity, direction relative to layering. Axelgold, directions relative to present horizontal. The precisions of (1) and (2) do not differ significantly. (3) Nilkitkwa Formation, Hazelton Group, Bait Range 55.6°N, 126.4°W, cleaning 550–600 °C, two sites reversed five sites normal polarity. (4) Telkwa Formation, Hazelton Group, Driftwood Range 55.8°N, 126.6°W, cleaning 80 mT and 550 °C, all normal polarity. (5) Telkwa Formation, Hazelton Group, Red Creek–Two Lake Creek area 56.5°N, 126.8°W, cleaning 550–600 °C and 60 mT, all reversed. (6) Savage Mountain Formation, Asitka Peak 56.7°N, 126.4°W, cleaning 60 mT and 500–550 °C, all normal polarity. (7) Savage Mountain Formation, Sustut Peak 56.6°N, 126.5°W, cleaning 60 mT and 500–550 °C, all normal. Directions in (3) to (7) are relative to bedding. *Th*, thickness of the section sampled in metres; *B*, number of sites each given unit weight in the analysis; *n*, number of cores (two specimens per core studied); *D, I* the mean direction; *k*, Fisher's estimate of precision; α_{95}, the 95% circle of confidence[35]. The position of the north pole is given; λ latitude, ϕ longitude. Unit weight given to each sampling site in the analysis.

Cretaceous the Stikine terrain was located ~1,300 km south of its present position relative to North America (Fig. 1).

The agreement between the mean values is excellent, better than would be expected from the associated errors (Fig. 5). The errors in palaeolatitudes of the base-maps have been obtained by giving unit weight to each rock unit when calculating the statistics of the mean polar wander path for North America[12] rather than to each determination of the geomagnetic field, that is, to each sampling site, of which there may be many in an individual study. The combined error estimates of Fig. 5 may therefore be too large. To use a procedure (that of giving unit weight to each rock unit) that probably overestimates errors seems more prudent than to use a method (that of giving sites unit weight) that probably underestimates them.

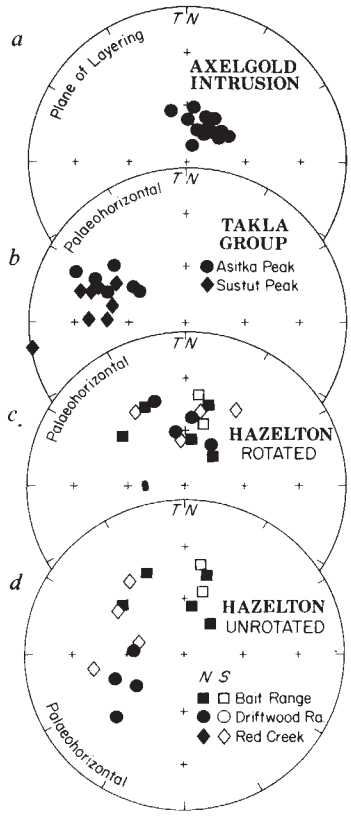


Fig. 4 Mean directions of characteristic magnetization at each collecting site. Directions are plotted on the lower hemisphere and the perimeter is the palaeohorizontal. The magnetization have normal polarity (north-seeking directions plotted) in the Axelgold and Takla rocks. In the Hazelton both polarities occur and the reversed directions (south-seeking directions, S) are shown by open symbols. In *d* Hazelton results are shown unrotated, correction being made for geological dip only. In *c* the directions are 'corrected' for rotations about local vertical axes (see text).

We suggest that the most probable time of motion of the Stikine terrain from its more southerly position was latest Cretaceous or early Tertiary. The apparent absence of post-Triassic palaeolatitudinal motion of the Alexander terrain relative to the craton[15] indicates that the Stikine block became wedged in behind the Alexander terrain at that time. The older limit for the time of this motion is set by the Axelgold intrusion whose age uncertainty has already been discussed. An important additional argument in favour of this late motion comes from considering the palaeomagnetic evidence from the Coast Plutonic Complex (CPC of Fig. 1). The Coast Plutonic Complex is a belt of Cretaceous and early Tertiary intrusions and metamorphic bodies that has not been disrupted by later deformation. The Stikine terrain is attached to the complex because it is intruded by them. Magnetic inclinations observed in Cretaceous intrusions within the Coast Plutonic Complex (such as the Howe Sound pluton[27]) are systematically too shallow, indicating that they were emplaced 10° or more south of their present position (see ref. 17, Fig. 3). Such data suffer from the absence of accurate corrections for possible post-emplacement tilt, but the effect is systematic over 400 km, and indicate a latest Cretaceous or earlier displacement consistent with our results.

A displacement of this magnitude, so late in the history of the Cordillera, and occurring so far to the east, raises the question of where the differential movement with stable North America is located. There is no geological evidence of such a late ocean closure between the localities sampled and the craton. The Cache Creek Group is thought to be ancestral Pacific Ocean floor which was thrust westward over the eastern margin of the Stikine terrain in post-Oxfordian time. Three major faults lie east of the localities sampled, and on two of them dextral strike-slip movement has been reported. The Pinchi Fault (Fig. 2) may have been a transform fault in the late Triassic, and right-lateral slip has probably recurred along it in the late Cretaceous and/or early Tertiary[28,29]. The Swannell Fault, situated to the east of the Pinchi, juxtaposes metamorphosed Proterozoic strata against later Palaeozoic and early Mesozoic volcanogenic rocks, and may mark the ancient cratonic margin at this point. No evidence of lateral slip has been reported from it. The northern Rocky Mountain Trench is the site of the third major fault (Fig. 2). Almost certainly right-lateral slip has occurred on it, but the amount

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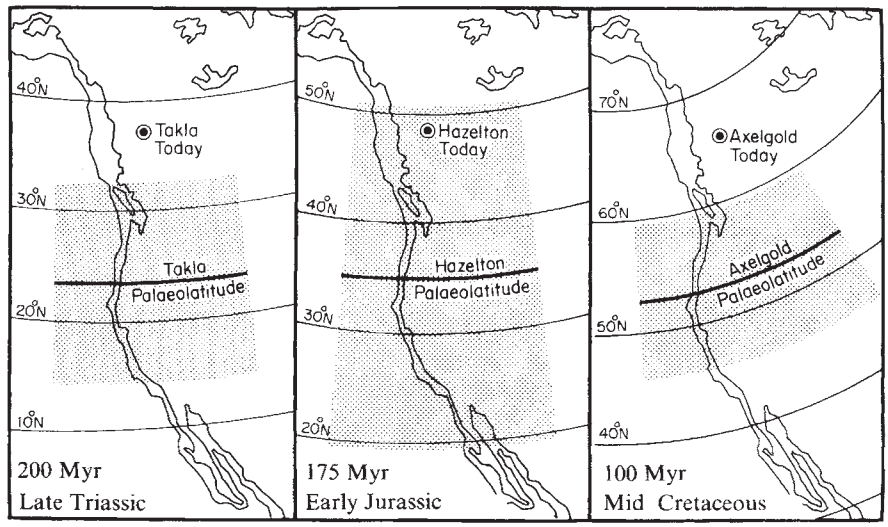


Fig. 5 Palaeolatitudes of the Stikine terrain compared with those calculated from cratonic North America. The reference palaeopoles[12] used for North America are for 200 Myr 68°N, 093°E, for 175 Myr 78°N, 110°E, for 100 Myr 69°N, 188E. The error zones ($P=0.05$) combine the errors in both base maps and palaeolatitude observations from the Stikine terrain. Error estimates are probably too large (see text).

is unknown. It probably extends northwards into the Yukon Territory as the Tintina Fault (Fig. 1). Between 350 and 500 km of right-lateral movements since the mid-Cretaceous has been reported from the Tintina Fault[30,31], and the displacement may, in part, continue down the northern Rocky Mountain Trench and, in part, diverge westwards down such faults as the Pinchi and the Frazer lineaments (*F*, Fig. 1). The motions that can be documented geologically are much less than the 1,300 km derived from our evidence. For reasons discussed elsewhere[32] Wrangellia was very probably attached to the Stikine terrain (Fig. 1) and rode northwards with it in the late Mesozoic or early Tertiary[32].

Our interpretation conflicts with that of Symons[33] who argues that the area "has not been tectonically rotated or translated relative to stable North America". His arguments are based on his palaeomagnetic results from the Topley intrusions of the Stikine terrain. These intrusions are Jurassic in age, probably somewhat younger than the Hazelton Group. The Topley region has been intruded and partially covered by early Tertiary igneous rocks which have yielded K–Ar ages of between 50 and 63 Myr, and there was extensive hydrothermal activity and mineralization of economic importance at this time. We believe that Symons has misinterpreted his data, which are in fact consistent with our own. Symons' results (ref. 33, Fig. 3 and Table 1) fall into two distinct groups (*A* and *B*) which we believe have two separate origins. *A* magnetizations occur at eight sites in four of the Topley intrusions. They have a high mean stability index ($SI=20$) (ref. 33), normal and reversed polarity, and a mean inclination of 47°, corresponding to a palaeolatitude of 30° indistinguishable from that we find for the Hazelton (Fig. 5). The declinations are scattered and range from 294 to 348°, indicating variable anticlockwise rotations among the four Topley intrusions similar to the rotations we find for the Hazelton. The *A* magnetizations are, therefore, very similar to the magnetization of the Hazelton Group, and we suggest that they are primary and represent the Jurassic geomagnetic field. *B* magnetizations occur at seven sites in three other Topley intrusions[33]. Their polarity is always normal, they have a low mean stability index ($SI=4$), and are well grouped with much steeper inclinations than *A* ($D=336°$, $I=74°$; $N=7$; $k=96$; $\alpha_{95}=6°$). The corresponding pole, 75°N, 176°W, is statistically identical with the mean pole for North America at 60 Myr (77°N, 173°W; $\alpha_{95}=7°$) (ref. 12). Hence, the less stable *B* magnetization was apparently acquired during the early Tertiary when igneous and hydrothermal activity was widespread in the Topley area. Symons averages all his results together and obtains a mean inclination of 63°, corresponding to a palaeolatitude of 40° ±11°, which is not significantly different from that expected from cratonic North America. We suggest that his average result has no tectonic significance because it is derived from combining a Jurassic magnetization (*A*) with a Tertiary magnetization (*B*).

Although the continuity of rock units indicates that the Stikine terrain has behaved as a coherent entity, it has not behaved as a rigid plate, because the declinations of the magnetizations directions are highly variable (Table 1). On its eastern margin the Axelgold Intrusion has rotated 63° clockwise relative to cratonic North America (rotations were calculated using equation (9.2) ref. 18). Clockwise rotations are very common in the western Cordillera, and have been ascribed to motions of small blocks activated by right-lateral shear of the Pacific and North American plates—the so-called 'ball-bearing' hypothesis[34]. However, our palaeomagnetic declinations show large, variable and commonly anticlockwise rotations; 43° and 62° for the two Takla localities, 41° and 103° for two of the Hazelton localities and 14° clockwise for the third. Of course, the observed western declinations could be caused by much larger clockwise rotation (360° less the above angles) and so bring their sense of rotation into line with those generally found in the western Cordillera. However, the simplest explanation is that different parts of the Stikine terrain have undergone variable amounts of predominantly anticlockwise rotation relative to cratonic North America. This is inconsistent with the simple ball-bearing hypothesis[34]. The Stikine terrain at least in its eastern part seems to be an assemblage of slices of crust of continental thickness with areal dimensions comparable to or less than the thickness of the lithosphere, and which are separated by faults with high strains (Fig. 2). Within each slice the strata are only gently tilted or folded[24].

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Different species of messenger RNA encode receptor and secretory IgM μ chains differing at their carboxy termini

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Biosynthetic studies in the presence of an inhibitor of glycosylation indicate that individual human lymphoma-derived cell lines can synthesize both membrane receptor and presumptive secretory forms of IgM μ chains. The receptor form has a larger polypeptide chain than the secretory form and possesses a different C-terminus, but similar N-terminus, consistent with the presence of a C-terminal hydrophobic 'tail' for integral membrane binding. Messenger RNA isolated from these cells directs the synthesis of both forms of μ chain in a wheat germ translation system, indicating the presence of independent mRNAs for each form. It is proposed that the synthetic pathways for receptor and secretory IgM diverge at the post-transcriptional level, possibly by differential RNA splicing to give mRNA molecules with or without a translatable 'tail' segment.

RECEPTOR immunoglobulin molecules on the surface of B lymphocytes behave in many respects like typical integral membrane proteins. By contrast, serum immunoglobulins, which function as effector antibodies, are hydrophilic globular proteins. It is reasonable to postulate that this dichotomy of function of immunoglobulins is dictated by a duality of structure. For instance, the immunoglobulin heavy chain may be expected to possess restricted regions of hydrophobicity which would function in membrane interaction. However, serum immunoglobulins do not seem to possess such hydrophobic regions. This observation has led previous workers to postulate the presence of a hydrophobic peptide on the receptor form of immunoglobulin, which is absent from the otherwise similar secretory form[1,2]. Alternative explanations have also considered the possibility that differences in glycosylation of the receptor form could cause conformational changes which would favour membrane binding[2].

There has been a steady accumulation of indirect evidence as to the manner of membrane binding by receptor immunoglobulin. The μ chain from radioiodinated splenocyte receptor IgM has been shown to be of higher molecular weight than its secreted counterpart by SDS-polyacrylamide gel electrophoresis (SDS-PAGE)[3]. We have made a similar observation for receptor IgM and IgG μ and γ chains synthesized in several human lymphoid cell lines[4], and a similar phenomenon has been reported for receptor IgM of a mouse lymphoid cell line[5]. Buoyant density and solubility measurements in the presence of non-ionic detergents have shown a greater hydrophobicity for receptor IgM, as opposed to secretory IgM[6,7]. In spite of these findings, the basis for the observed differences in receptor and secretory immunoglobulins remained unclear. The apparent molecular weight differences could be due either to variation in the extent of glycosylation or to a difference in polypeptide chain lengths. More recently, carboxypeptidase digestion and peptide mapping techniques have been used in several laboratories[5,8,9] to analyse membrane associated μ chains for a modified C-terminus. These studies have so far yielded conflicting results, which, in some experiments, may be due to contamination by the secretory μ-chain species.

A separate but related question to the structure of receptor immunoglobulin concerns the biosynthetic pathway which distinguishes it from secretory immunoglobulin. Previous models[10,11] have proposed that divergence of the pathways occurs just before secretion, based on the finding that intracellular transport of both forms is very similar[12,13]. However, investigations of the events preceding synthesis of the polypeptide chains have not been previously reported.

In this article we report on the structure and biosynthesis of receptor IgM chains in human lymphoid cells representing various stages of B-lymphocyte differentiation and immunoglobulin expression. We have used the drug tunicamycin as an inhibitor of glycosylation. Size comparisons of non-glycosylated μ chains indicate that the membrane form has a larger polypeptide chain than the secretory form. The larger membrane form of μ chain was isolated and shown by carboxypeptidase digestion to have a modified C-terminus (but a normal N-terminal sequence) relative to secretory μ chain, consistent with the presence on receptor μ chain of a C-terminal peptide, possibly hydrophobic in nature, capable of interaction with the plasma membrane. Unexpectedly, two forms of μ chain were detected intracellularly in Daudi and Raji nonsecretory cells, one resembling the membrane form and one resembling the secretory form with the former predominating in Daudi cells and the latter form predominating in Raji cells. This same pattern of μ-chain expression was found by cell-free translation of Daudi and Raji mRNA, indicating that the two forms of μ chain can be

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